Freephones - A look at free-to-use payphone projects via PhilTel

Agenda

- Who are we?
- What's a payphone?
- What happened to them?
- What are "freephones"?
- What's PhilTel?
- What have we learned?
- Future Plans

Who are You?

Mike

- Software engineer
- Philly 2600 (2012-)
- FreePBX/Asterisk user (2012-)
- Networks of Philly (2018-)
- Dialup.world (2021-)

Naveen

- Computer engineer, IT
- Telephone collector, phone phreak (2016-)
- Asterisk user (2018-)
- Payphone inventory (2019-)
- PhreakNet (2021-)
- Asterisk/DAHDI developer and consultant (2021-)
- BBS software developer (2022-)
- Environmental activist (2016-)

What is a Payphone?



What is a Payphone?





What happened to them?

Take a tour of Detroit's last surviving pay phones

All over Detroit, we found battered pay phones in various states of disrepair. It's much harder to find a dial tone.

JC Reindl Detroit Free Press

NATIONAL

Last call: New York City bids an official farewell to its last public pay phone

MAY 24, 2022 · 11:47 AM ET

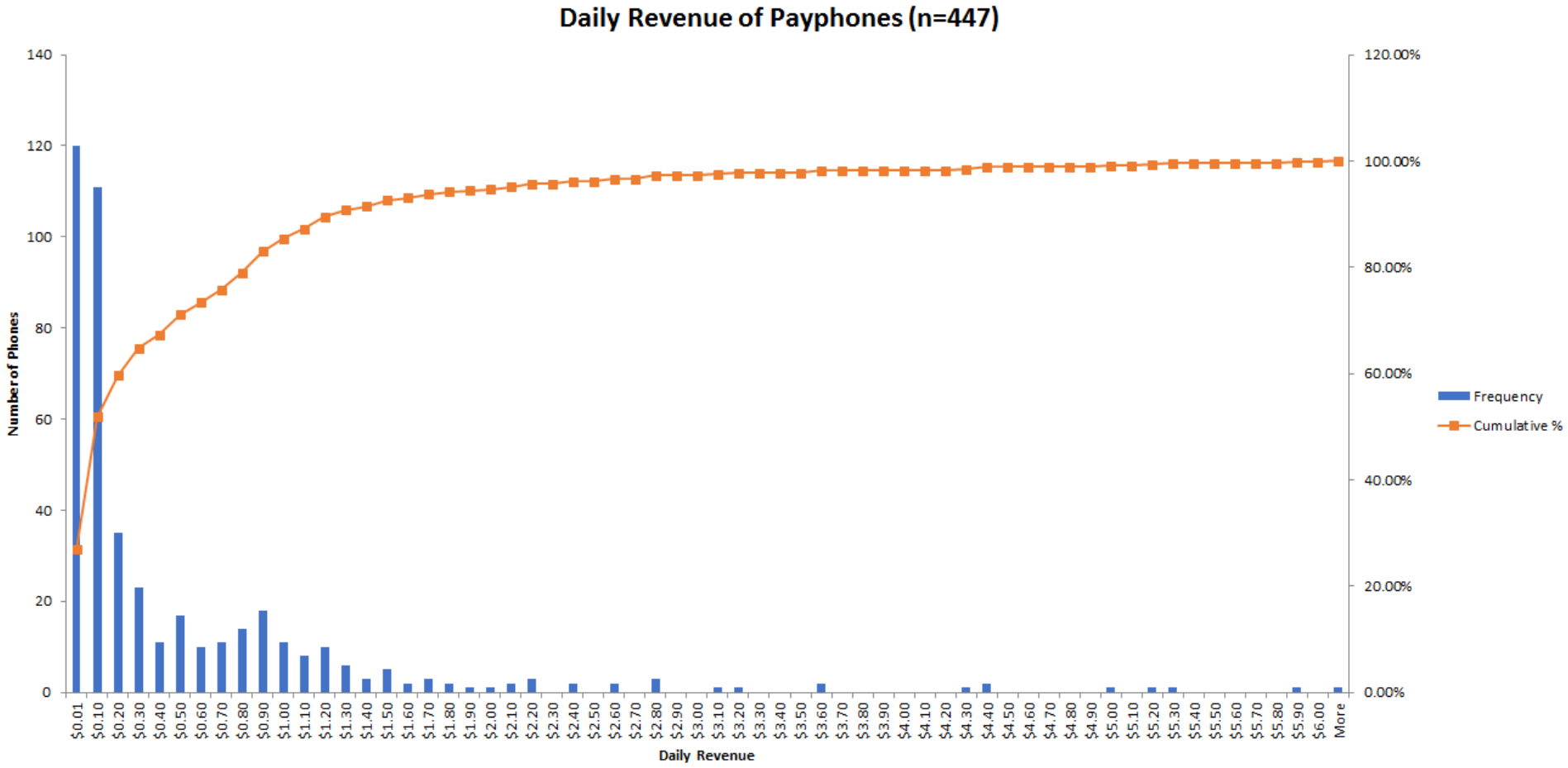
By Rachel Treisman



Searching for Cleveland's last payphone, I found something better

Updated: Mar. 16, 2026, 7:21 a.m. | Published: Mar. 16, 2026, 6:30 a.m.

What happened to them?



What happened to them?

- **98%** of adults in the USA have a cell phone (Pew Research)
- Philadelphia population as of July 1 2024 is **1,573,916** (census.gov)
- **78.8%** of Philadelphians are 18+ (census.gov)
- (1573916*.788)*.02 = **~24,808** Philadelphia adults without a cell phone
- Xfinity Mobile Arena max seated capacity: **~21,000**

What happened to them?



What are Freephones?

- Essentially, payphones but **free-to-use**
- Public-use, courtesy phones
- Often older payphones or armored phones because they are rugged, vandal-proof, and weather resistant
- Anyone can set up a freephone, you don’t need to be a telephone company
- Often utilize VoIP for phone service, off-the-shelf hardware, and open-source software
- May have unique services and easter-eggs, including operator services, connection to collectors’ networks, etc.

What are Freephones?

- The first freephone project is likely **Futel**, based in Portland Oregon, started in 2014 by Karl Anderson and Elijah St. Clair
- Started with one phone in Karl's front yard, now has 17 phones in 3 states



What are Freephones?

- Freephone Projects (that we know of)
  - Futel - Portland, OR
  - **PhilTel – Philadelphia, PA**
  - RandTel - Central VT
  - Good Phone Project – Monroe County, NY
  - 0Dial – Canada
  - OIRCT - France

What is PhilTel?

- **PhilTel** is inspired by Futel, but based in **Philadelphia**
- Bring payphones back but make them free-to-use this time because payphones are cool
- Use as much discarded and second-hand stuff as we can, build with COTS hardware and open-source software
- Large upfront cost (>$300), small recurring cost (<$10/mo)
- Benefit people without cell phones, preserve public telephony



The First Phone





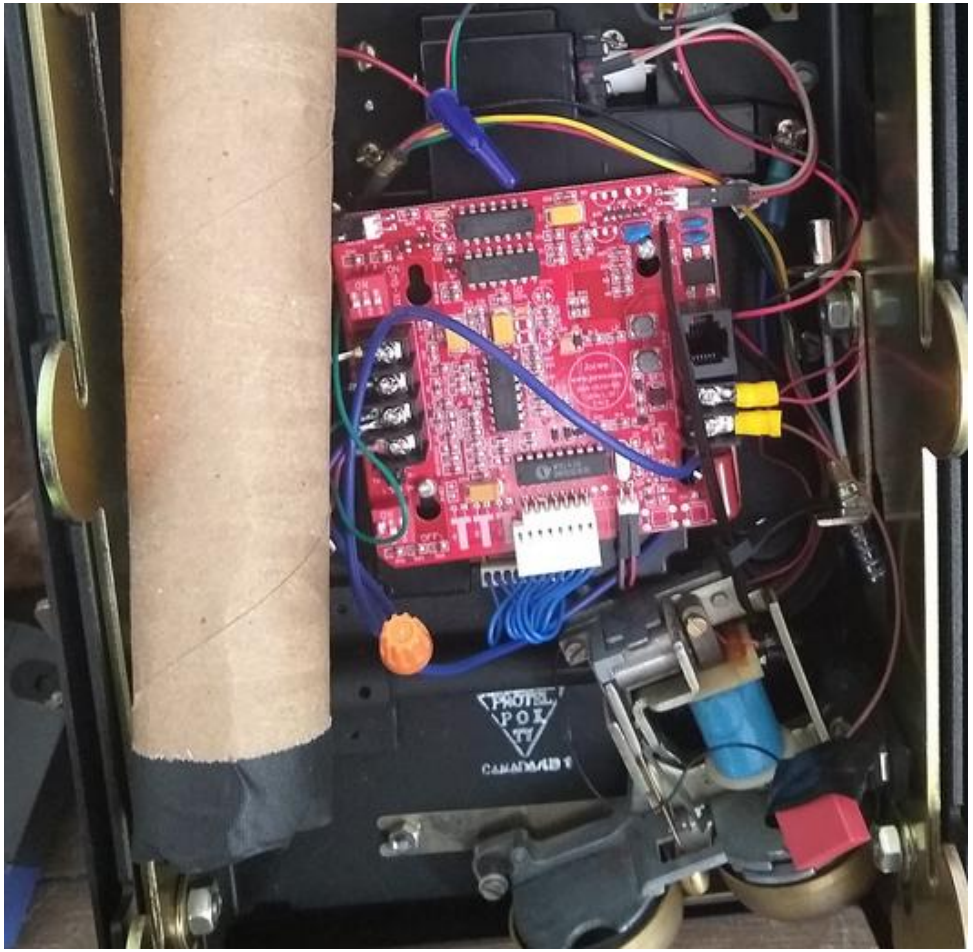


2110V Coinless Board

- “Dumb” phone board made for armored phones
- Acts as an (almost) drop-in replacement board for most GTE-style payphones
- No longer made (but still available)
- Good solution if you don’t want to mess with Protel hardware/software



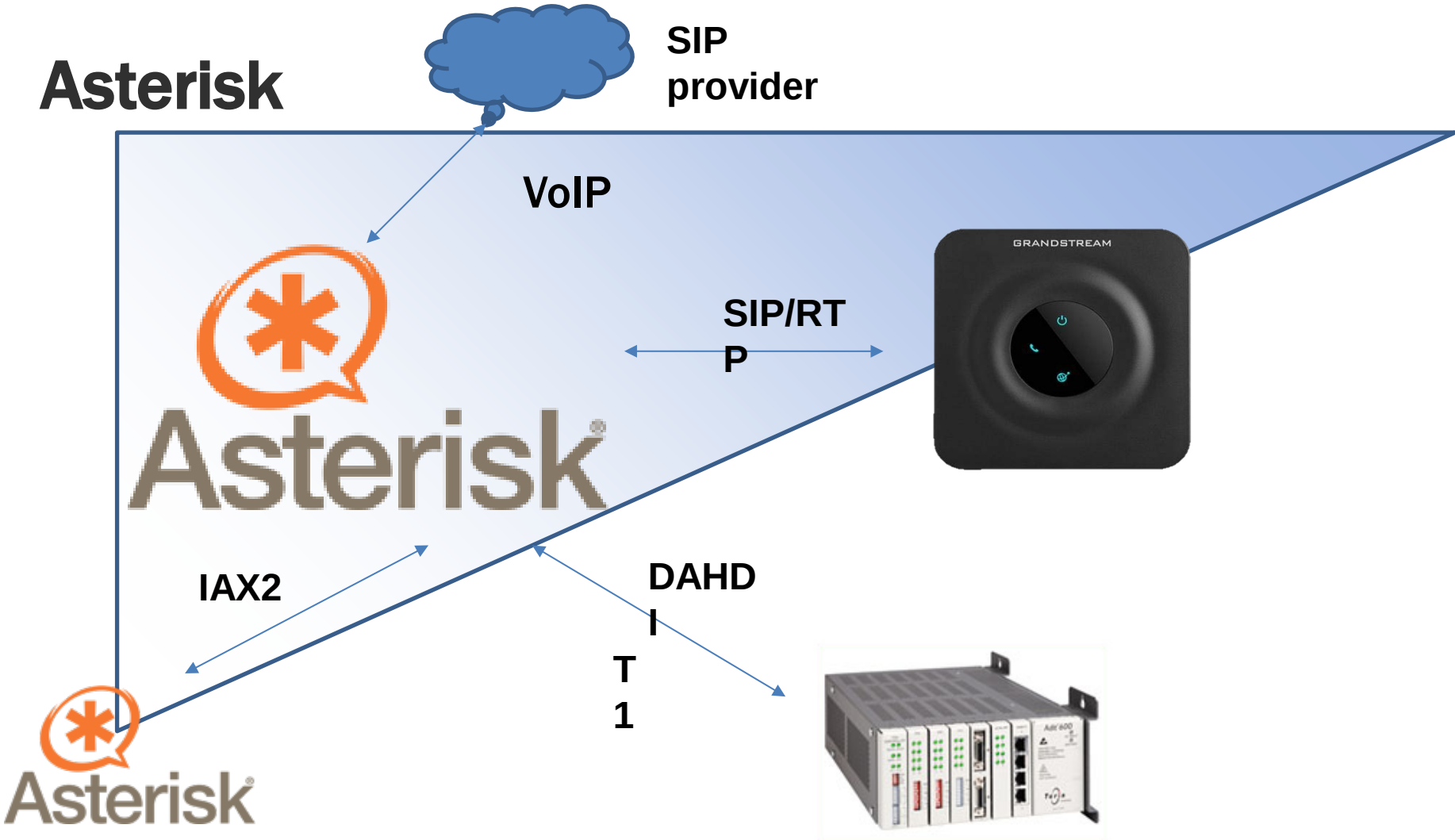
2110V Coinless Board (Continued)



Grandstream HT801 ATA

- Analog Telephone Adapter
- Allows analog telephones to speak SIP (Session Initiation Protocol)
- Registers as a client to our telephone exchange so we can make/take calls
- Remotely monitored with GDMS (Grandstream Device Management System) for free!




Asterisk
SIP
provider
VoIP
Asterisk
SIP/RT
P
GRANDSTREAM
IAX2
DAHD
I
T
1
Asterisk

Our Portable Proof of Concept





Flagship Phone @ IffyBooks

- Initially Installed December 17, 2022, reinstalled several times since.
- Had a party, hackers and phreakers showed up





People seemed to like it

NEWS > PHILADELPHIA

Philadelphians welcomed PhilTel's first free public phone as a small way to resist big tech

PhilTel's founders hope the phone installed Saturday is the first of many and will help restore the city's "public communications infrastructure."



npr

10 PHILADELPHIA

The Washington Post Sign in

Remember the pay phone? This man is bringing them back at no cost.

'There are lots of people who can't afford a cellphone or a phone bill,' PhilTel co-founder Mike Dank said



By Cathy Free

December 16, 2022 at 6:00 a.m. EST



CNN US Audio Live TV Log In

A volunteer collective is setting up free pay phones in Philadelphia

By Zoe Sottile, CNN

Published 10:04 AM EST, Sun December 4, 2022



Courtesy Mike Dank

Technical.ly

A Second Phone





What Did We Learn?



What Did We Learn?

- VPNing the ATA can work, but SIPS/SRTP works better
  - ISPs can block or traffic-shape VPN traffic
  - VPN means…
    - More interoperability challenges
    - More latency
    - More maintenance (expiring certificates)

What Did We Learn?

- Less hardware is better
  - Originally tried to run everything through a small router
  - More cost/maintenance when not needed for all installations
  - Hardware limitations for WiFi



What Did We Learn?

- ATAs work until they don’t, and most aren’t great
- Grandstream ATAs are still the best we know of
  - Plagued with firmware issues for V1s initially
  - We have had 2 die on one phone
  - One died during a demo
  - V2 ATAs have been nothing but trouble



What Did We Learn?

- People will try their best to ruin a good thing
  - Repeated vandalism on the 2nd phone
  - Don’t prank call 911
    - Have a demo mode for exhibits
- There are good people too!

What Did We Learn?

- Some parts are often meant to be replaced, not repaired
- Don’t spend all your time trying to fix something that doesn’t want to be fixed
  - You may not be able to trust it



What Did We Learn?

- Accidents happen
- This is a marathon, not a sprint



What Did We Learn?



What Did We Learn?

- People love to play with and are nostalgic for payphones

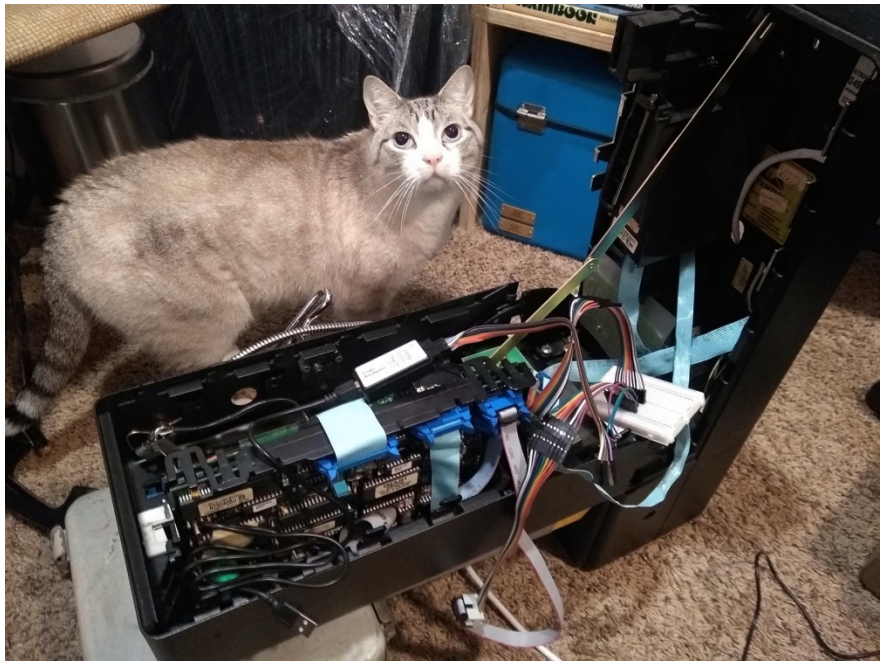


What Did We Learn?

- Write everything down
  - Good documentation shows your work is reproducible (and helps those starting out)
  - You'll use it yourself when you forget in a few months
  - We keep a lot of documentation
    - https://philtel.org/
    - https://github.com/philtelco/philtel-docs
    - https://payphone.club/ (work in progress)

What Did We Learn?

- Community is everything





What Did We Learn?

- Don’t let someone tell you something is impossible



Questions?

Mike

- mike@philtel.org
- Famicoman (Libera IRC)
- https://famicoman.com
- https://github.com/Famicoman

Naveen

- interlinked@philtel.org
- InterLinked (Libera IRC)
- https://interlinked.us
- https://github.com/InterLinked1

https://philtel.org/contact/

https://philtel.org/donate/

https://github.com/philtelco/philtel-presentations/